

Araz Mehrabani

Mechanical Engineer | Wind Farm Planning · windPRO · QGIS

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📅 Date of birth: 28.07.1989



PROFILE & CORE COMPETENCIES

Mechanical engineer focused on **wind-farm planning, site screening and technical project engineering**. Hands-on windPRO/QGIS workflows from GIS constraints and wind resource through layout/wakes/AEP to noise, shadow flicker, curtailment and technical reporting.

TECHNICAL FOCUS

windPRO: METEO, WAsP, PARK, OPTIMIZE, DECIBEL, SHADOW; wind data/resource, turbine layout, wake losses, AEP and layout optimization.

Constraints & Environment: receptors, minimum spacing, noise/shadow checks, operating modes, bat curtailment and realistic energy-loss assessment.

QGIS & Site Screening: raster/vector, Shapefile/GeoTIFF, buffer, clip/intersect, distance analysis, roughness, contours, protected areas, grid/road access.

Data & Reporting: Excel (Pivot, Power Query, lookup, charts), Power BI (DAX, dashboards), Python/CSV processing and technical reports.

PROFESSIONAL EXPERIENCE

Development Engineer — Sabaniroo, Tehran 01/2022–02/2023

- Performed **QGIS** site screening in southeastern Iran using raster/vector layers, buffer/clip/intersect and distance analysis for wind potential, protected areas, grid connection, road access and site accessibility.

Mechanical Engineer — TehranRigzar, Tehran 03/2017–12/2021

- Designed and technically supported conveyors, jaw crushers and sand washers; coordinated with fabrication/workshop teams on welded structures, dimensions and tolerances.

Working Student – Mechanical Design — Ario Fidar Maham Co., Tehran 11/2012–05/2013

- Recreated a 40x90 jaw crusher in **CATIA V5**, including individual components, assemblies and manufacturing drawings.

SELECTED WIND PROJECTS

Advanced Wind Farm Planning – Hochschule Flensburg — End-to-end workflow with METEO/WAsP/PARK/OPTIMIZE/DECIBEL/SHA roughness/contours/receptors, wind resource, turbine siting, wake losses & AEP, layout optimization, noise/shadow and curtailment assessment.

20 MW Offshore Wind Turbine – interdisciplinary development project — Coordinated within the Loads & Dynamics team with tower, foundation and other development teams on load interfaces, input data and technical boundary conditions.

EDUCATION

M.Sc. Wind Energy Engineering, Hochschule Flensburg — current grade: 1.9 03/2023–11/2026

M.Sc. Mechanical Engineering, Hakim Sabzevari University — Control & Robotics 09/2014–01/2017

B.Sc. Mechanical Engineering, Islamic Azad University — Structural Dynamics 09/2008–06/2013

LANGUAGES

German: B2 (actively improving). English: C1 / IELTS 7.0. Persian: Native.